MeSAM: Multiscale Enhanced Segment Anything Model for Optical Remote Sensing Images

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***Abstract*— Segment anything model (SAM) has been widely applied to various downstream tasks for its excellent performance and generalization capability. However, SAM exhibits three limitations related to remote sensing (RS) semantic segmentation task: 1) the image encoders excessively lose high-frequency information, such as object boundaries and textures, resulting in rough segmentation masks; 2) due to being trained on natural images, SAM faces difficulty in accurately recognizing objects with large-scale variations and uneven distribution in RS images; and 3) the output tokens used for mask prediction are trained on natural images and not applicable to RS image segmentation. In this article, we explore an efficient paradigm for applying SAM to the semantic segmentation of RS images. Furthermore, we propose multiscale enhanced SAM (MeSAM), a new SAM fine-tuning method more suitable for RS images to adapt it to semantic segmentation tasks. Our method first introduces an inception mixer into the image encoder to effectively preserve high-frequency features. Second, by designing a mask decoder with RS correction and incorporating multiscale connections, we make up the difference in SAM from natural images to RS images. Experimental results demonstrated that our method significantly improves the segmentation accuracy of SAM for RS images, outperforming some state-of-the-art (SOTA) methods. The code will be available at https://github.com/Magic-lem/MeSAM.**

***Index Terms*— High-frequency, multiscale, remote sensing (RS), segment anything model (SAM), semantic segmentation.**

I. Introduction

REMOTE sensing (RS) semantic segmentation assigns a clear category to each pixel in the image to achieve accurate classification of surface objects at the pixel level. It plays an important role in urban planning, resource management, agricultural applications, and disaster monitoring [1], [2], [3].

To address the challenges of complex backgrounds and diverse categories in RS semantic segmentation, many specialized neural network models, such as multi-attention-network (MANet) [4] and local-global class-aware network (LOGCAN) [5], have been proposed to effectively enhance the accuracy of object segmentation in RS images. However, designing domain-specific neural network models entails a considerable amount of domain knowledge and network engineering. Besides, facing different datasets, these dedicated models always need to be retrained from scratch due to poor generalization. Simultaneously, the diversity of RS images inevitably poses a significant challenge to the generalization ability of the models [6], [7]. Therefore, these dedicated models usually spend substantial computational resources to be retrained on different datasets.

To overcome the above problem, large-scale visual models have been proposed as universal models by pretraining them on considerable data and then fine-tuning for segmentation. Some methods, e.g., segment everything with a generalist painter (SegGPT) [8], pretrained the foundation model on natural images by mix-context training for general purposes (i.e., many different tasks) and then fine-tuned it for semantic segmentation. However, methods based on this paradigm usually achieve limited performance because of the difference between natural and RS images. Therefore, some large-scale RS models [9], [10] have recently been proposed to address the problem. Nevertheless, these RS large-scale models mainly emphasize pretraining of the foundation model for general purposes rather than the specific segmentation task, also limiting the segmentation performance. Noteworthy, the model based on the former paradigm of pretraining on natural images for general purposes has this limitation, too.

Therefore, to avoid the aforementioned limitations, we introduce a paradigm based on pretraining on natural images for segmentation tasks and then fine-tune it for RS images. Although it is not a new paradigm for small-scale models, there is still a lack of a method based on this paradigm for large-scale models for RS semantic segmentation. That is because two challenges limit the adaptation of large-scale natural image segmentation models, particularly segment anything model (SAM) [11], for semantic segmentation of RS images. First, proven by Wang et al. [12], the backbone of many large-scale visual models (i.e., the vision transformer) has a character of low-pass filtering, which can lead to unclear

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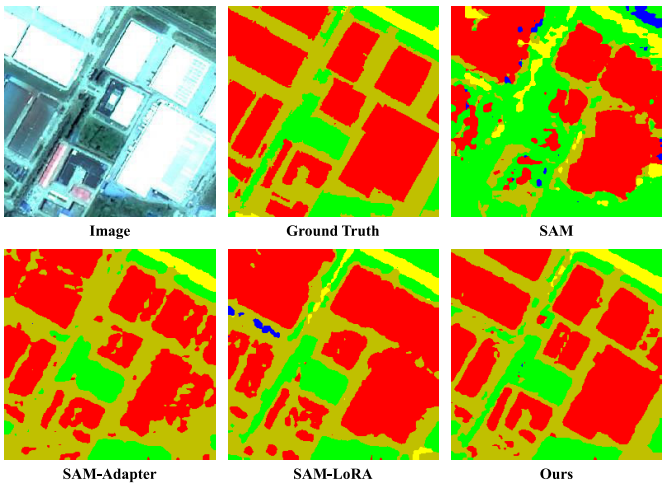


Fig. 1. Visualization of the segmentation masks of SAM and our methods. The prediction header in the mask decoder of SAM is modified to output semantic categories. Noteworthy, the segmentation map of SAM has no semantic.

segmentation edges. Second, the characteristics of natural and RS images are different. Indeed, RS images have lots of objects of different scales with large variations, while the scale variation in objects in natural images is usually much smaller. Thus, large-scale natural image segmentation models cannot adapt well to the diverse object scales and uneven distribution in RS images.

Especially for SAM, in addition to the above limitations, the output tokens in it are trained for natural images, which is inappropriate for semantic segmentation of RS images due to the difference in distribution between natural and RS images. RS-Prompter [13] and Osco et al. [14] solved this problem by adding an extra prompt generator and inputting prompts of RS images, respectively. Hence, prompt datasets are indispensable for their methods, whereas making prompt datasets is time-consuming and costs lots of resources. Furthermore, the mechanisms in [13] and [14] have been introduced for RS instance segmentation rather than semantic segmentation.

To alleviate the above issues, we propose a multiscale enhanced SAM (MeSAM) for semantic segmentation of RS images. Specifically, SAM [11] is adopted as the foundation model in our method for RS semantic segmentation. Thanks to huge training data with masks, SAM was pretrained by prompt learning for segmentation tasks rather than self-supervised learning for general purposes [15], which endows the model of impressive generalization for segmentation. On the other hand, although SAM's excellent generalization capabilities allow it to directly segment RS images, its performance is limited due to the aforementioned issues, as shown in Fig. 1. Therefore, we first introduce an inception mixer to rebuild the attention module in the SAM image encoder, forming inception transformers, thereby avoiding loss of high-frequency resulting from low-pass filtering of vision transformers. Afterward, we present the multiscale features extraction via multiscale convolutional kernel and skip connections from the inception transformers, thus adapting the original SAM to RS images with objects of variant scales. Finally, by correcting output tokens for natural images in the decoder with an extra learnable RS token and a correction module, we can mitigate the limited performance resulting from different distributions between natural and RS images.

The contributions of this article can be summarized as follows.

1) We propose a large-scale model for RS image semantic segmentation, the so-called MeSAM, which shows better performance than the compared state-of-the-art (SOTA) methods for RS semantic segmentation.
2) An efficient fine-tuning paradigm with the enhancement of high-frequency and multiscale features is proposed for SAM, filling the gap of large-scale semantic segmentation of pretraining on segmentation and then featuring on RS images. Compared with SOTA-efficient fine-tuning methods, such as low-rank adaptation (LoRA) [16] and adapter [17], the proposed fine-tuning methods achieve much better results (see Section IV-D1).
3) We introduce an inception mixer to fine-tune the image encoder in SAM for RS segmentation, improving the ability to extract high-frequency and multiscale features for semantic segmentation, which improves the performance for objects at different scales and with rich textures or edges.
4) We design a learnable token and correction module in the masking decoder of SAM for RS semantic segmentation, which can calibrate the output tokens for natural images to RS images without any prompt fine-tuning. Benefiting from that, we can leave out the collection of prompt datasets for RS semantic segmentation.

The rest of this article is organized as follows. Section II provides a brief review of related works. Section III presents a detailed description of SAM's application paradigm and the proposed MeSAM for RS image semantic segmentation. Comprehensive quantitative and qualitative experimental results and analyses are given in Section IV. Finally, Section V draws the conclusions of this work.

II. Related Works

A. Remote Sensing Image Semantic Segmentation Models

With the successful application of deep learning in the field of computer vision (CV), deep-learning-based semantic segmentation of RS images has attracted widespread attention. In the early stages, neural network models that achieved success in CV, such as fully convolutional networks (FCN) [18] and SegNet [19], were directly applied to RS image segmentation tasks. However, the complexity and diversity of RS images force researchers to continually design new neural network models. For instance, there are models such as hybrid dilated convolution U-net (HDCUNet) [20] for segmenting coastal aquaculture areas in RS images, separable residual SegNet (SR-SegNet) [21] for water body segmentation, and UNet-like transformer (UNetFormer) [22] and multi-scale skip connected and asymmetric-convolution-based U-Net (MACUNet) [23] designed for fine-resolution remotely sensed image segmentation.

In recent years, vision transformer [24] have also been exploited in RS for image segmentation. However, since

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the transformer adopts a global self-attention mechanism, it tends to overlook spatial information within each patch [25]. Therefore, most transformer-based RS image segmentation models combine convolutional neural networks (CNNs) with vision transformers to fully use both local and global information [26], enhancing segmentation performance on RS images. By designing an adaptive fusion module and combining transformer with CNNs, Gao et al. [25] extracted coarse-grained and fine-grained feature representations of various semantic scales at different stages. In [27], a hybrid deep neural network has been implemented for aerial image segmentation by combining a Swin-transformer-based encoder with a CNN-based U-shaped decoder. Although the above-mentioned work has achieved high performance, transformer-based segmentation models usually have huge computational costs. To address this issue, Wang et al. [22] used a lightweight CNN as encoder and developed an efficient global–local attention mechanism to build a transformer module, effectively reducing computational complexity.

B. Large-Scale Remote Sensing Models

Although transformer-based segmentation technology performs well in RS images, most models often do not have a strong generalization ability, and retraining them on various datasets typically requires significant computational burden. With the significant progress of large-scale foundation models in natural language processing (NLP) and CV tasks [28], [29], [30], foundation models in the field of RS have the potential to solve this problem [9], [10], [31], [32], [33]. Wang et al. [9] first explored large-scale visual models suitable for RS tasks and built a vision transformer with approximately 100 million parameters, achieving competitive performance across various RS tasks. To investigate the impact of model parameter count, Cha et al. [10] introduced the first billion-level foundation model in the RS domain, demonstrating the effectiveness of increasing parameters. However, most RS foundation models mainly focus on a single modality without temporal and geo-context modeling, which limits their ability to perform different tasks. Guo et al. [31] proposed a billion-level multimodal RS foundation model, which takes the time series of optical and synthetic aperture radar (SAR) data as input and can perceive multimodal time RS imagery.

Many studies [15], [34], [35] consider building pretraining frameworks more suitable for RS images. Addressing the complexity and density of objects in RS scenes, Sun et al. [15] specifically designed a foundation model training method for the RS domain based on generative self-supervised learning, achieving SOTA results on multiple datasets. Based on masked autoencoder (MAE), Cong et al. [34] proposed a pretraining framework for temporal or multispectral satellite imagery, demonstrating the effectiveness of pretraining transformers on satellite images. Subsequently, Reed et al. [35] proposed a pretraining method that can explicitly learn the relationship between data at different scales, which can more accurately capture features in RS images.

The scale of RS datasets has always been an important factor affecting the performance of RS foundation models. To solve this problem, Mendieta et al. [36] constructed a compact but diverse dataset from multiple sources and established a continual pretraining pipeline to simultaneously use the powerful representations of ImageNet to guide and accelerate learning, producing strong performance with minimized overhead. Bastani et al. [37] combined Sentinel-2 and NAIP images and proposed an RS dataset, which improved the performance of basic models on downstream tasks.

Nowadays, in addition to improving the performance of foundation models on downstream tasks, some studies also explored more lightweight foundation models [38]. Moreover, some multimodal RS foundation models [39], [40], [41] that integrate text and other modal data have shown excellent performance in various multimodal tasks. Although pretraining models on RS images could be a solution to obtain a good generalization ability, limitations are still evident due to the heterogeneity among downstream tasks [42].

C. Segment Anything Model

SAM [11] is a visual foundation model trained on over a billion masks, exhibiting robust zero-shot generalization capabilities and, therefore, widely used in various downstream tasks [43], [44]. However, the original segmentation results of SAM lack semantic information, and studies have found that in specific domains, such as medical and RS, SAM cannot compete with domain-specific models trained from scratch [17], [45], [46]. Consequently, many works [47], [48], [49] guided SAM to accomplish specific task segmentation by introducing additional prompts. de Oliveira et al. [46] evaluated the role of various prompt types in medical image segmentation and suggested using boxes as prompts in the initial stage and points as prompts in subsequent refinements to guide SAM segmentation, achieving zero-shot segmentation performance that equals or surpasses the existing methods. In the field of RS images, RSPrompter [13] successfully applied SAM to instance segmentation tasks by autonomously generating prompt embeddings. However, this method has limitations in the feature extraction capability for RS images, as it only modifies the prompts input of SAM.

Furthermore, some works [14], [17], [50], [51] enhance the segmentation capabilities of SAM on specific datasets by fine-tuning. While introducing the LoRA, Zhang and Liu [52] modified the number of output masks in SAM so that it can directly output semantic information, achieving end-to-end medical image semantic segmentation based on SAM. In addition to the mentioned research, many improvement methods have been applied to SAM, aiming to obtain more precise segmentation results for complex objects. It is worth mentioning that Ke et al. [53] designed a learnable high-quality output token, which is inputted into the decoder of SAM to obtain higher quality segmentation masks. It can be pointed out that compared with other fields, there is a large gap in the application of SAM in RS image segmentation.

D. High-Frequency Information Processing

Due to the outstanding performance of vision transformer (ViT) in extracting global information, it has exceled in many

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CV tasks. However, Wang et al. [12] demonstrated that the self-attention mechanism is essentially a low-pass filter. The image features processed by multiple layers of ViT lose a considerable amount of high-frequency information, including edges and textures. To address this issue, some approaches, from the perspective of frequency-domain analysis, use methods, such as Fourier transform [54], Laplace transform [55], and wavelet transform [56], to enhance and extract high-frequency information. Si et al. [57] proposed a mixer based on channel segmentation, combining the low-frequency information from the transformer with the high-frequency information from convolutional layers. Pan et al. [58] proposed an efficient attention mechanism to reduce the additional computational cost due to the transformer. They divided the multihead attention layers into two groups, focusing on low-frequency and high-frequency information, respectively. Similarly, scattering vision transformer (SVT) [59] proposed the use of the dual-tree complex wavelet transform (DTCWT) to decompose image features and effectively reduce computational complexity through the Einstein blending method (EBM) technique. In this article, we explore some solutions for SAM to effectively extract high-frequency information from RS images (e.g., object textures and edges), which plays a crucial role in semantic segmentation.

III. Methodology

In this section, we will introduce the paradigm of fine-tuning SAM for RS image semantic segmentation, as well as our proposed method. We will include the following aspects: a brief review of SAM, the implementation of end-to-end semantic segmentation in SAM by introducing fine-tuning modules and adjusting the mask output, and MeSAM for better RS image semantic segmentation based on high-frequency and multiscale feature enhancement.

A. SAM: A Review

SAM is a visual foundation model trained on more than 11 million images and one billion masks for segmentation, demonstrating strong zero-shot generalization capabilities. It achieves effective segmentation of arbitrary objects by simple interactive prompts. SAM mainly consists of three components: an image encoder, a prompt encoder, and a mask decoder. The image encoder of SAM is built on the ViT pretrained with MAE [60], with a 14×14 window of attention and four equidistant global attention blocks. The prompt encoder processes various types of prompts—sparse prompts (points, boxes, text) and dense prompts (masks)—encoding them before inputting them into the mask decoder for the final mask prediction. The mask decoder comprises modified transformer blocks and a dynamic mask prediction head, mapping inputs, such as image embedding and tokens, to the final mask.

B. Fine-Tuning SAM for Remote Sensing Image Segmentation

Since most foundation models are more general-purpose rather than focused on a specific task, their performance in semantic segmentation is limited. Therefore, this section mainly explores a paradigm for fine-tuning the segmentation foundation model, SAM, to the task of semantic segmentation of RS images.

To enable SAM to output relevant semantic information without relying on prompts or additional work, we adjust the output mask quantity of the mask decoder to align it with the pixel category count in semantic segmentation, ensuring that each mask corresponds to each category, to achieve end-to-end semantic segmentation. As a result, in this paradigm, the model can directly obtain semantic masks without any prompt information. Therefore, we remove the prompt encoder to reduce model parameters and computation.

Moreover, to overcome the limitations of SAM in extracting features from RS images while ensuring that the original structure in SAM is not destroyed, in addition to MeSAM, as shown in Fig. 2, we also introduce the universal fine-tuning modules Adapter [17], [61] and LoRA [16] into the image encoder of SAM to get the SAM-Adapter and SAM-LoRA methods for RS image segmentation.

1) SAM-Adapter: Fig. 2(a) illustrates the SAM structure based on the Adapter fine-tuning module. The Adapter module introduces additional task-specific knowledge, such as texture or frequency information, using appropriate prompts. Consistent with [17], these prompts are automatically generated based on input features. Specifically, the prompt input P_i of the ith layer is added to the output of the transformer after two layers of multilayer perceptron (MLP) processing and one layer of activation function. Thus, we have

$$F_{i+1} = O_i + M_2(\sigma(M_1(P_i))) \tag{1}$$

where O_i is the output feature of the ith layer, F_{i+1} refers to the input feature of the next layer with respect to the ith layer, M_1 and M_2 refer to two MLP layers, respectively, and σ is the Gaussian error linear unit (GELU) activation function.

In this method, the original weights of the image encoder of SAM are frozen. Only the newly added Adapter modules need to be fine-tuned. To enhance the processing capabilities of features, we simultaneously fine-tune the entire mask decoder, not just the mask prediction head. Thanks to the lightweight design of the decoder in SAM, this step does not lead to an excessive computational burden.

2) SAM-LoRA: Fig. 2(b) depicts the SAM structure based on the LoRA fine-tuning module. Similar to SAM-Adapter, SAM-LoRA introduces additional layers between two consecutive transformer layers. The difference is that the LoRA module implements skip connections through low-rank fully connected layers, without introducing additional information. The architecture and positions of LoRA modules are shown in Fig. 2(b). Consistently with the work [52], we apply the LoRA layer to the query and value projection layers in the transformer.

During fine-tuning, the original weights of all the transformers remain frozen, and only the fully connected layers in the newly added LoRA modules require fine-tuning.

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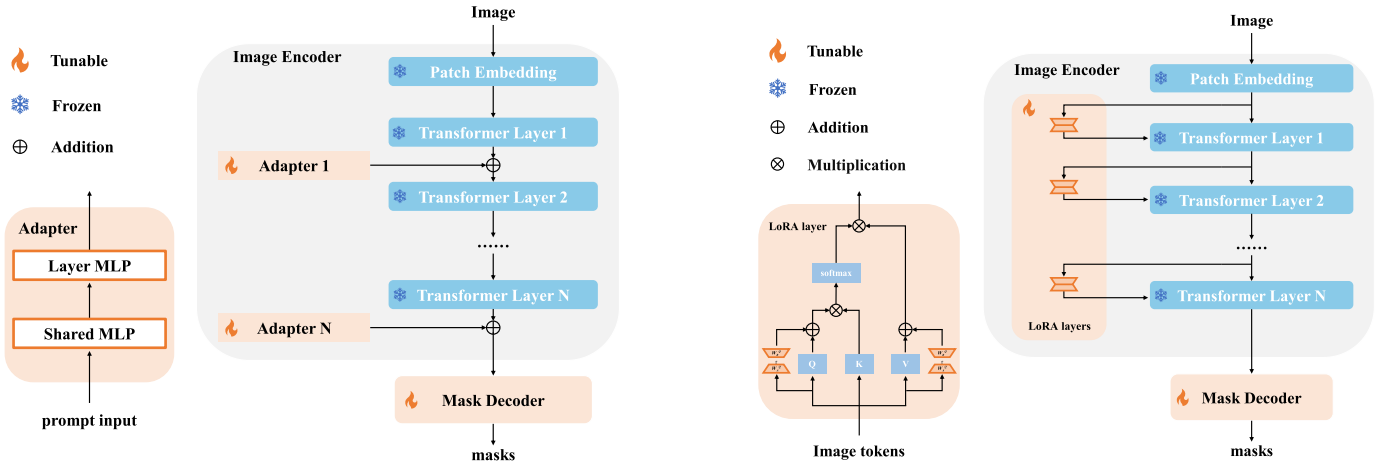


Fig. 2. Model structure of SAM with Adapter and LoRA modules. (a) Adapter modules are introduced between two consecutive layers of the image encoder in SAM. During fine-tuning, the weights of all the transformers in the encoder are frozen, and the weights of the mask decoder and Adapter modules are adjusted together. (b) LoRA modules are added to the query and value projection layers in each transformer of the image encoder. All the newly added modules and the mask decoder in SAM are fine-tuned together.

C. MeSAM

1) Overview: Given an RS image $X \in \mathbb{R}^{C\times H\times W}$, where C represents the channels, and $H \times W$ denotes the image size, the goal of semantic segmentation is to predict the category of each pixel in the spatial dimension, i.e., the segmentation mask $M \in \mathbb{R}^{H\times W}$. Each pixel may belong to any of the classes y_i, where $i \in (0, 1, \ldots, n)$ and n is the number of categories. y_0 indicates that the pixel belongs to the background. The overall structure of the proposed MeSAM is shown in Fig. 3. The input image X is encoded by the image encoder and is mapped to the segmentation mask M in the mask decoder. Thus, we have

$$M = \mathcal{D}(\mathcal{E}(X), T) \tag{2}$$

where $\mathcal{E}$ represents the image encoder, $\mathcal{D}$ refers to the mask decoder, and T represents the tokens input to the mask decoder.

To prevent ViT from losing too many high-frequency features of the image, we introduce the Inception Transformer [57] to reconstruct the image encoder. We exploit convolutional layers and maximum filtering to compensate for the loss of high-frequency information. Furthermore, in layers with global attention, we used larger kernel sizes in the Inception Transformer to adapt it to the object size diversity of RS images.

Regarding the mask decoder, to overcome the unsuitability of output tokens for RS image segmentation, we introduce the additional RS output (RS-Output) token instead of fine-tuning them. The advantage of this method is that it can make full use of the powerful performance of SAM and improve segmentation accuracy. Meanwhile, to better exploit feature information at different scales and enhance the segmentation performance of image details, we designed a multiscale connection to extract the output of the transformer layer with global attention and input it into the decoder as early features. The output of the last layer of the encoder is fused to obtain the final image encoding outcome.

2) Image Encoder With Inception Transformer: To improve the performance of SAM in RS image segmentation, we use the Inception Transformer to replace the original ViT. The structure of the Inception Transformer is shown in Fig. 4. Compared with the traditional transformer structure, it improves the ability to extract high-frequency information by introducing an Inception Mixer to modify the multihead self-attention (MSA) layers. The structure of the Inception Mixer is shown in Fig. 4, which includes 3×3 and 5×5 structures. The input feature $F \in \mathbb{R}^{N\times C}$ is first split along the channel dimension into a low-frequency component, $F_l \in \mathbb{R}^{N\times C_l}$, and a high-frequency component, $F_h \in \mathbb{R}^{N\times C_h}$. The low-frequency component, F_l, is processed by the original MSA of SAM to obtain Y_l, calculated as follows:

$$Y_l = \text{MSA}(F_l). \tag{3}$$

The high-frequency component, F_h, is further divided into $F_{h1} \in \mathbb{R}^{N\times(C_h/2)}$ and $F_{h2} \in \mathbb{R}^{N\times(C_h/2)}$. F_{h1} is processed by maximum filtering [i.e., MaxFilter$(\cdot)$] and pointwise convolution [i.e., PWConv$(\cdot)$] to obtain Y_{h1}, and Y_{h2} is obtained after processing F_{h2} using pointwise convolution and depthwise convolution [i.e., DWConv$(\cdot)$]. This is because maximum filtering can emphasize important features in local areas, while convolution operations help in detecting edges and in enhancing high-frequency information. Afterward, the inception mixer combines the three components, Y_{h1}, Y_{h2}, and Y_l, to obtain Y_m as follows:

$$Y_{h1} = \text{PWConv}(\text{MaxFilter}(F_{h1})) \tag{4}$$

$$Y_{h2} = \text{DWConv}(\text{PWConv}(F_{h2})) \tag{5}$$

$$Y_m = \text{Fusion}(\text{Concat}(Y_{h1}, Y_{h2}, Y_l)) + F \tag{6}$$

where Fusion$(\cdot)$, achieved by a depthwise convolutional layer, refers to the step of further merging all the features and Concat$(\cdot)$ is the concatenation operator. After the feedforward network (FFN) at the end, the final output of the Inception Transformer, Y, can be expressed as

$$Y = Y_m + \text{MLP}(Y_m) \tag{7}$$

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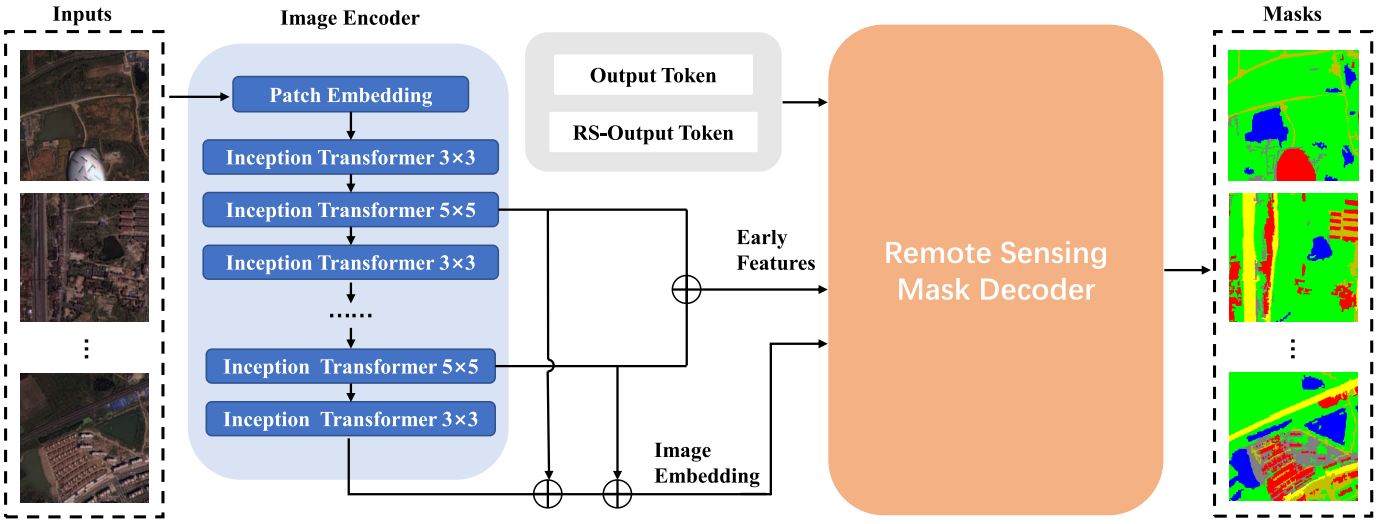


Fig. 3. Overall structure of MeSAM. After the features of the RS images are extracted by the image encoder, they are input to the decoder for processing in a multiscale connection manner and mapped to the semantic segmentation masks.

where MLP(·) refers to an MLP processing. For sake of simplicity, the normalization terms in all the formulas are omitted.

As shown in Fig. 4, we only fine-tune a few additional layers in the inception mixer to effectively reduce computational costs and preserve the original information and feature extraction ability of SAM. By disabling the added modules, the original effectiveness of the SAM encoder can be restored.

3) Mask Decoder With Remote Sensing Correction and Multiscale Connection:

a) Remote sensing correction: Due to the differences between natural images and RS images, we introduce a learnable RS-Output token to adjust the tokens in input to the mask decoder, as shown in Fig. 5, whose size is consistent with the output token. After concatenating with the SAM's output tokens, the tokens are processed through two decoder blocks and then inputted to the dynamic mask mapping layer. In particular, unlike the output token, the RS-Output token is processed by the same cross-attention layer and then spatially multiplied with the fused features through the new independent MLP layer, as shown by the red arrow in Fig. 5, which can capture feature information at different scales in RS images and correct the original masks to improve segmentation performance.

b) Feature fusion and multiscale connection: The input features of the decoder are divided into two parts: the early features and the final output features of the encoder. In our method, early features are extracted from all the layers with global attention in the image encoder, not limited to only the first global attention layer [53]. This is because compared with natural images, RS images are more diverse and complex in terms of object scales, making it difficult to fully extract all the information using only single-scale features. As shown in Fig. 5, the inception mixer output features of all the global attention layers are input into the decoder as early features after weighted fusion. Moreover, when facing semantic segmentation tasks, it is difficult for SAM to extract a large amount of semantic information from the final output of the encoder. To enhance the multiscale semantic information in the final encoded features, we also fuse the output features of all the global attention layers into the final output of the encoder. The early features input into the decoder are first fused with the image embedding for predicting corrected masks. Specifically, consistently with [53], we initially exploited transposed convolution to upsample the two features and then sum them elementwise to obtain the fused features.

In MeSAM, since the mask decoder of SAM is lightweight, we only freeze the two decoder blocks in the mask decoder to improve the processing and mapping capabilities of features. The cross-attention layer and transposed convolution layer are also fine-tuned along with the new modules.

IV. Experiments

This section will introduce the detailed configuration of the experiments and the results obtained on the dataset. First, we will compare our results with the SOTA methods. Next, we will analyze the efficiency of the proposed method and its effectiveness in extracting high-frequency information. Finally, we will validate the performance of all the components in our method through the ablation study.

A. Dataset

To validate the effectiveness of the proposed method, we conducted evaluations on three publicly available RS semantic segmentation datasets with different spatial resolutions. They include two high-resolution datasets (i.e., WHDLD and DLRSD [62], [63]), as well as a low-resolution dataset (i.e., Sentinel-2 Cloud Mask Catalogue [64]).

1) DLRSD: This dataset is sourced from the UC Merced archive, comprising a total of 2100 RGB aerial images with a spatial resolution of 0.3 m. Each image has a size of 256 × 256 pixels, covering pixel distributions for 17 semantic classes, including airplane, bare-soil, buildings, cars, chaparral court, dock, field, grass, mobile-home, pavement, sand, sea,

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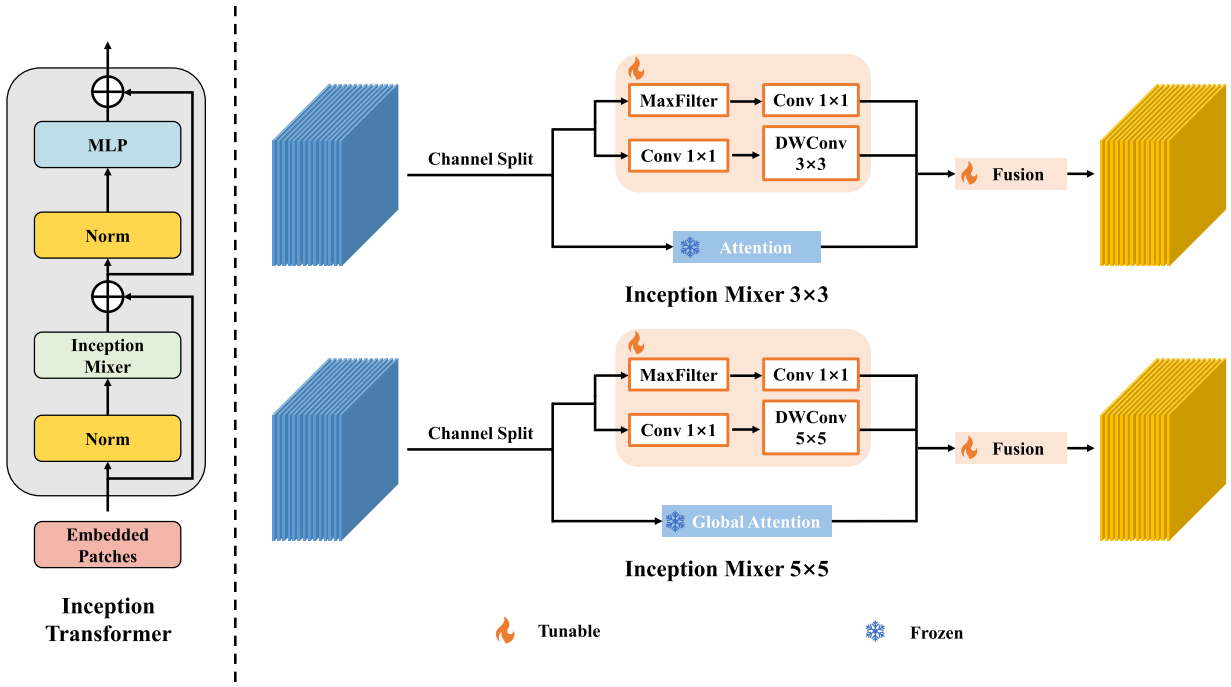


Fig. 4. Structure of Inception Transformer. Inception Transformer replaces the MSA with the inception mixer, which consists of MSA and an additional high-frequency information enhancement branch. During fine-tuning, only the weights in the additional branch need to be updated.

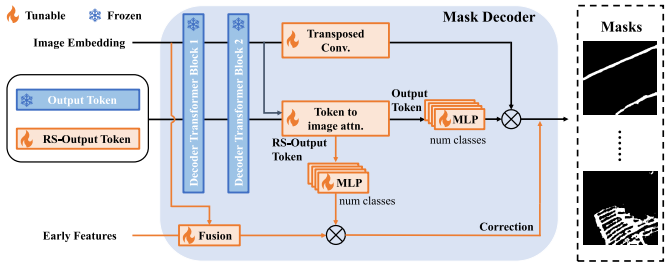


Fig. 5. Structure of the mask decoder. Each MLP layer corresponds to the segmentation mask of one class.

ship, tanks, trees, and water. The class distribution in this dataset is shown in Table I. In the experiments, we randomly divided 90% of the dataset for training and the remaining 10% for testing. Furthermore, the training set has been randomly divided into a 9:1 ratio for training and validation purposes.

2) WHDLD: This dataset is a multilabel dataset released by Wuhan University in 2018. It contains 4940 RGB images from Gaofen-1 (GF-1) and Ziyuan-3 (ZY-3) satellites, with each image size of 256 × 256 and a spatial resolution of 2 m. The dataset includes six classes: bare soil, building, pavement, road, vegetation, and water, and the class distribution is shown in Table II. For this dataset, we performed a random split with a ratio of 9:1 for training and testing. Similarly, 10% of the training set has been randomly used for validation.

3) Sentinel-2 Cloud Mask Catalogue: The Sentinel-2 Cloud Mask Catalogue dataset contains 513 subscenes of 1022 × 1022 pixels each with a resolution of 20 m, randomly sampled from the 2018 Level-1C Sentinel-2 archive. It consists of 13 channels corresponding to 13 spectral bands, including three categories: clear, cloud, and cloud_shadow. The class distribution is shown in Table III. In our experiments, we only used three bands as input: red, green, and blue (B4, B3, and B2). We randomly cropped the images to a size of

TABLE I
CLASS DISTRIBUTION FOR THE DLRSD DATASET

Dataset	Class	Images	Pixels	Proportion(%)
DLRSD	airplane	100	448066	0.3
	bare-soil	754	14957125	11
	building	713	13979440	10.2
	cars	897	3650472	2.7
	chaparral	116	3026039	2.2
	court	105	1789456	1.3
	dock	100	908702	0.7
	field	103	6406008	4.7
	grass	977	20581511	15.1
	mobile-home	102	2507209	1.8
	pavement	1331	34230567	25.1
	sand	291	4506658	3.3
	sea	101	4225935	3.1
	ship	103	2015667	1.5
	tanks	100	121537	0.1
	trees	1021	17590688	12.8
	water	208	5600520	4.1

TABLE II
CLASS DISTRIBUTION FOR THE WHDLD DATASET

Dataset	Class	Images	Pixels	Proportion(%)
WHDLD	bare soil	3722	13076295	4.1
	building	3162	35902158	11.1
	pavement	3881	37297258	11.5
	road	4631	13945297	4.3
	vegetation	3539	144809306	44.7
	water	3886	78717526	24.3

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TABLE III
CLASS DISTRIBUTION FOR THE SENTINEL-2 CLOUD MASK CATALOGUE DATASET

Dataset	Class	Images	Pixels	Proportion(%)
Sentinel-2 Cloud Mask Catalogue	clear	2051	247202446	45.9
	cloud	2052	282829624	52.6
	cloud_shadow	2051	7887418	1.5

512 × 512, obtaining a total of 2052 images. We randomly divided the dataset into a 9:1 ratio for training and testing, where 10% of the training set has been randomly used for validation.

B. Evaluation Metrics

We use three metrics to evaluate the performance of all the segmentation methods, including overall accuracy (OA), average F1-score (F1), and mean intersection over union (mIoU). We define the calculation of each metric using true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs).

OA refers to the proportion of samples correctly classified by the model across all the categories, calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{8}$$

Furthermore, to comprehensively assess the model's segmentation performance across different categories, we calculate the mIoU and F1-score for each segmentation method, as follows:

$$\mathrm{mIoU} = \frac{1}{k}\sum_{i=1}^{k}\frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{9}$$

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{10}$$

where precision and recall are defined as

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad \mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{11}$$

C. Implementation Details

In this article, we use SAM with ViT-B as the backbone and AdamW as the optimizer with a weight decay value of 0.01. The initial learning rate is set to $6e^{-4}$ and decayed during training using a cosine annealing strategy. We specify a maximum of 100 training epochs and use cross-entropy loss as the loss function, with a batch size of 16 for both the DLRSD and WHDLD datasets. All the experiments are implemented with the PyTorch framework on a single NVIDIA GTX 3090 GPU.

For SAM based on ViT-B, the numbers of layers with global attention are 2, 5, 8, and 11, respectively. Therefore, in our method, we use the Inception Transformer with a kernel size of 5 × 5 on these specific layers, while on the other layers, a kernel size of 3 × 3 Inception Transformer is used.

D. Results and Discussion

1) Comparison With State-of-the-Art: To validate the effectiveness of the proposed method, we use SegNet [19], attentive bilateral contextual network (ABCNet) [65], bilateral awareness network (BANet) [66], MANet [4], UNetFormer [22], MACUNet [23], fully transformer-based UNet-like transformer (FTUNetFormer) [22], LOGCAN [5], and pretrained foundation models such as Scale-MAE [35], geospatial foundation model (GFM) [36], and SatlasNet [37] as benchmarks for comparison. Meanwhile, we also verified the performance of frozen SAM, SAM fine-tuned with Adapter and LoRA, as well as end-to-end fine-tuned SAM on the two datasets.

As shown in Table IV, our proposed method outperforms the current SOTA works on both the datasets. Specifically, the experimental results on the DLRSD dataset demonstrate that compared with the suboptimal advanced model LOGCAN, our method improves the OA, F1-score, and mIoU by 1.10%, 0.75%, and 1.16%, respectively. In contrast, methods based on pretrained foundation models performed unsatisfactorily on the DLRSD dataset, with SatlasNet achieving the highest OA of 81.31%, F1-score of 78.56%, and mIoU of 65.77%. These scores are approximately 4%, 7%, and 10% lower than ours.

Meanwhile, the performance of our method significantly surpasses other approaches based on SAM. Specifically, in terms of OA, our method achieves a score of 85.37%, which is higher than SAM-Frozen, SAM-Adapter, and SAM-LoRA by 9.74%, 2.65% and 2.56%, respectively. In terms of the F1-score, our model gets a score of 85.46%, which is 3.43% higher than SAM-LoRA. Regarding the crucial mIoU metric, our method scores 75.28%, which is much better than the value obtained by SAM-LoRA, i.e., 70.27%, and the one obtained by SAM-Adapter, i.e., 70.18%, with improvements of over 5%. Due to the limited size of the DLRSD dataset, consisting of only 2100 images with an imbalanced class distribution, the SAM-Full method, characterized by a vast number of updated parameters, struggles to converge effectively. These results not only demonstrate the advancement of our method in RS image processing but also validate its adaptability and efficiency for complex semantic understanding.

The visualization results of all the methods on the DLRSD dataset are shown in Fig. 6. It can be intuitively seen that the segmentation effect of our method is significantly better than other methods, which is consistent with the results in Table IV. Looking at the results in the first row, our method's predictions on the "pavement" class are closest to the ground-truth mask, while other methods tend to mistakenly predict it as "buildings." Furthermore, the results in the third and fifth rows demonstrate that our method achieves superior segmentation performance in dealing with dense and small objects, and it is also closer to the ground-truth mask in regions with blurred boundaries.

In the WHDLD dataset, our method achieved a score of 84.49% in terms of OA. Compared with the advanced works LOGCAN and MACUNet, our method improved by 0.23% and 0.17%, respectively. Compared with SAM-

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TABLE IV

QUANTITATIVE COMPARISON WITH SOTA MODELS ON THE DLRSD AND WHDLD DATASETS. THE BEST VALUE IN A COLUMN IS SHOWN IN BOLD. * INDICATES THAT ONLY THE LEARNABLE PARAMETERS AT THE FINE-TUNING STAGE ARE REPORTED

Method	Backbone	DLRSD			WHDLD			Params (M)	Infer. Speed (ms / image)
		OA(%)	F1(%)	mIoU(%)	OA(%)	F1(%)	mIoU(%)		
SegNet	-	77.33	74.08	60.53	81.67	72.42	59.10	29.44	10.21
ABCNet	-	78.30	74.49	61.16	81.87	72.59	59.27	13.70	16.56
BANet	-	77.70	74.10	63.41	82.77	73.78	60.72	12.75	18.16
MANet	-	79.58	78.25	65.64	83.82	75.74	63.04	35.86	29.43
UNetFormer	-	78.42	77.81	64.85	83.53	75.98	63.17	11.73	19.74
MACUNet	-	80.72	77.55	64.94	84.32	76.09	63.48	5.15	36.49
FTUNetFormer	-	79.88	79.64	67.23	83.78	76.30	63.58	36.17	30.70
LOGCAN	-	84.28	84.71	74.11	84.26	76.87	64.37	30.91	39.07
Scale-MAE*	ViT-Large	71.48	65.17	50.80	78.68	69.82	55.77	4.33	29.93
GFM*	Swin-Base	68.49	61.08	46.65	79.26	70.36	56.55	4.25	36.58
SatlasNet*	Swin-Base	81.31	78.56	65.77	81.11	72.83	59.47	4.19	53.61
SAM-Frozen*	ViT-Base	75.63	74.84	61.36	72.93	60.00	46.18	3.08	32.68
SAM-Adapter*	ViT-Base	82.72	81.88	70.18	82.30	74.26	61.07	4.55	35.25
SAM-LoRA*	ViT-Base	82.81	82.02	70.27	82.51	74.31	61.18	4.63	36.37
SAM-Full*	ViT-Base	69.58	62.58	47.92	83.24	75.22	62.34	91.15	32.68
Ours*	ViT-Base	**85.37**	**85.45**	**75.27**	**84.49**	**77.16**	**64.63**	14.85	56.82

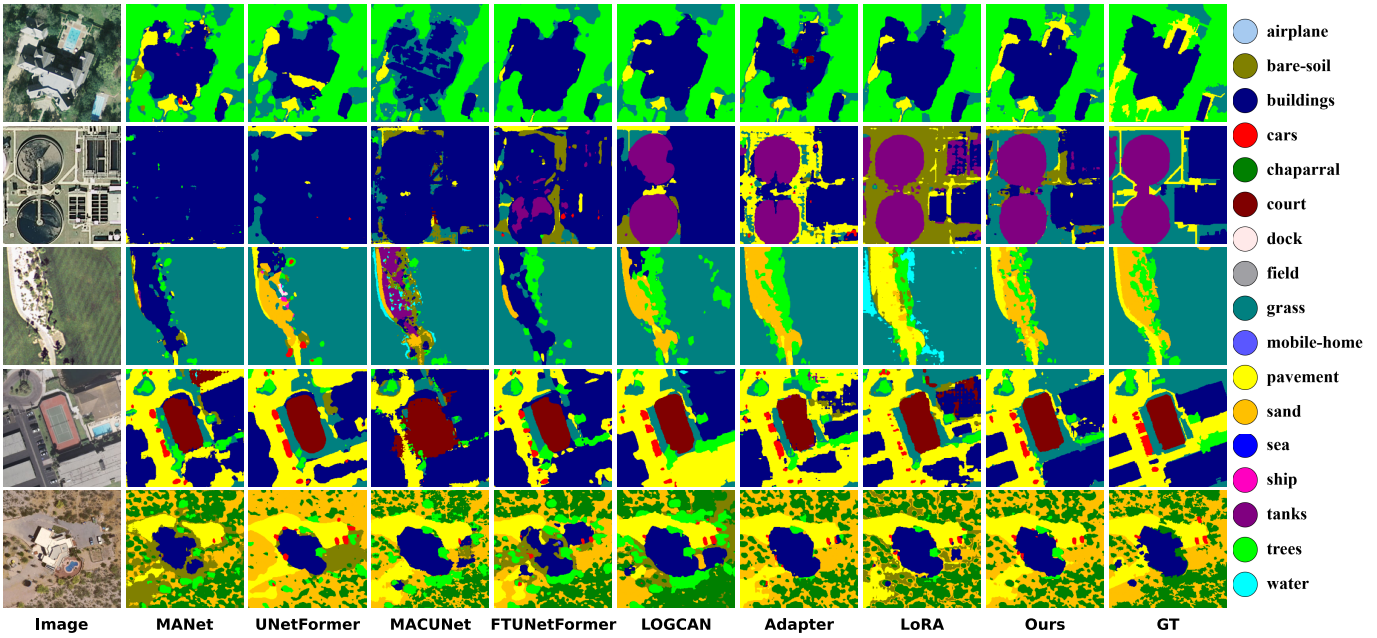


Fig. 6. Qualitative comparison of the visualization results of our method with other top-five methods and SAM-based methods on the DLRSD dataset.

LoRA, our method improved by nearly 2%. The proposed model outperformed all other models with a score of 77.16% for the F1-score. In comparison to methods based on pretrained foundation models, our approach outperforms Scale-MAE, GFM, and SatlasNet by 8.86%, 8.08%, and 5.16%, respectively, in terms of mIoU, and also exhibits a clear advantage in other metrics.

In addition, our model improved by 2.9% and 2.8% compared with SAM-Adapter and SAM-LoRA considering the F1-score, respectively. Regarding mIoU, our model performed outstandingly with a score of 64.63%, far exceeding SAM-LoRA's 61.18% and SAM-Adapter's 61.07%. The SAM-Frozen method fails to effectively apply to the WHDLD dataset for segmentation tasks, achieving only 72.93% in OA, 60.00% in F1-score, and 46.18% in mIoU, which is significantly lower than other methods. Furthermore, compared with SAM-Full, our approach only fine-tunes 16% of the parameters yet outperforms it, with respective improvements of 1.25%, 1.94%, and 2.29% across the three metrics. These results demonstrate that our model has a significant performance advantage in semantic segmentation tasks of RS images compared with classical fine-tuning methods, even surpassing the current SOTA models.

The visual results for the WHDLD dataset are shown in Fig. 7. MeSAM achieves better segmentation performance on dense and blurred boundary objects, especially in the "building" class, which is significantly better than other methods. When the object distribution is clustered, making it difficult to distinguish, the compared approaches tend to predict the area as belonging to the same category, whereas our method exhibits higher sensitivity to boundaries.

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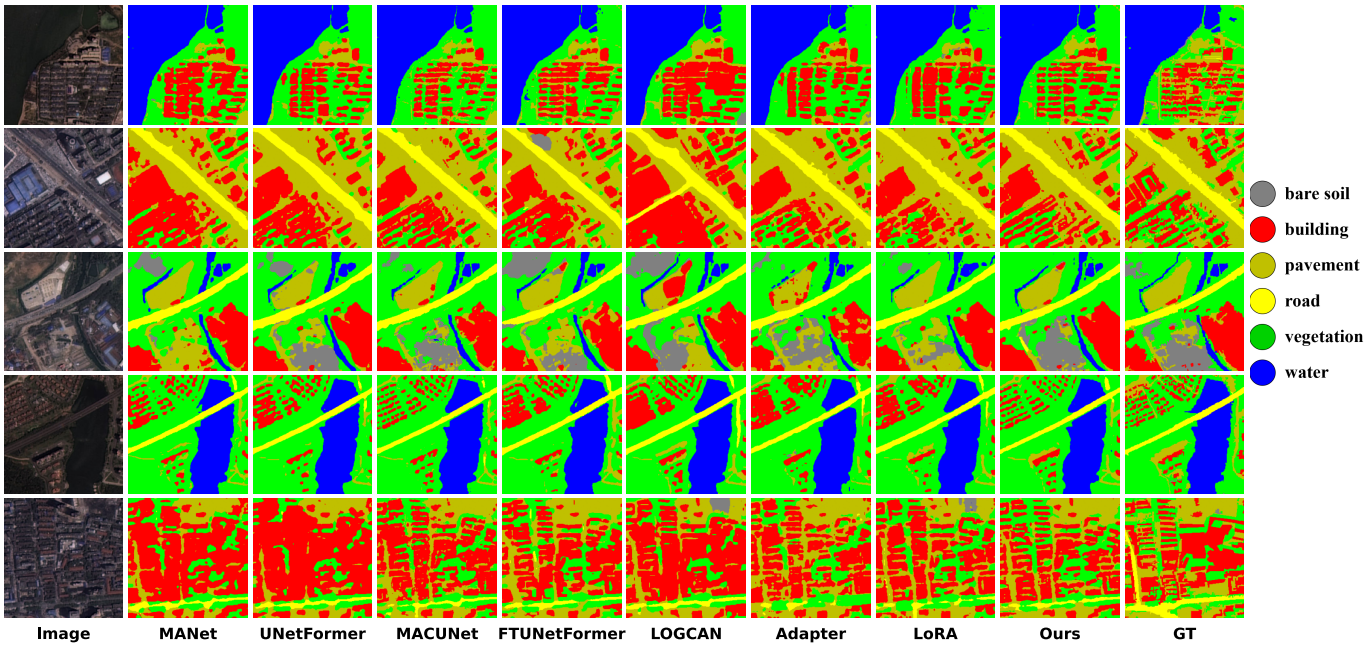


Fig. 7. Qualitative comparison of the visualization results of our method with other top-five methods and SAM-based methods on the WHDLD dataset.

TABLE V
IoU (%) per Class on the DLRSD Dataset. The Best Value in a Column Is Shown in Bold and the Second-Best Value Is Underlined

Method	air.	bare soil	bui.	cars	cha.	court	dock	field	grass	mob. home	pav.	sand	sea	ship	tanks	trees	water	**avg.**
SegNet	44.80	43.15	59.93	64.55	65.05	70.43	18.31	87.32	57.99	58.57	70.93	58.99	78.53	61.25	46.45	65.38	77.27	60.53
ABCNet	37.86	48.19	53.85	61.86	59.23	53.87	39.28	**97.83**	61.71	56.52	69.23	62.82	92.78	63.64	31.32	67.99	81.68	61.16
BANet	50.12	48.66	56.71	62.44	68.57	50.36	48.60	97.82	60.66	57.50	69.87	66.52	92.33	66.85	32.70	67.64	80.66	63.41
MANet	54.08	47.12	60.02	67.92	67.36	66.53	52.31	97.78	61.66	59.45	72.35	64.61	93.22	66.08	34.26	67.46	83.63	65.64
UNetFormer	60.91	46.94	53.56	65.97	62.85	64.21	55.45	97.58	60.85	59.31	69.56	53.53	87.30	72.50	40.26	66.95	84.65	64.85
MACUNet	43.79	51.12	62.01	63.91	**71.26**	66.66	39.78	**97.83**	63.76	57.48	73.33	63.08	92.35	65.18	36.92	70.23	85.30	64.94
FTUNetFormer	60.66	47.63	56.63	66.71	66.89	71.77	55.02	97.77	62.78	56.36	72.18	67.93	92.63	70.24	46.49	67.55	83.60	67.23
LOGCAN	**69.96**	50.80	73.85	**73.13**	66.60	**79.27**	61.47	91.69	63.37	68.98	79.86	**73.79**	91.82	72.03	**82.78**	73.32	87.11	74.11
Scale-MAE	26.13	31.73	39.21	50.21	43.87	54.08	26.14	95.52	56.00	46.34	61.93	43.52	86.51	55.19	17.88	63.34	63.03	50.80
GFM	12.99	29.75	47.21	60.52	50.13	26.83	38.76	93.69	40.33	39.32	61.80	26.03	62.83	65.87	18.66	55.74	62.55	46.65
SatlasNet	55.72	54.03	63.13	62.11	66.85	73.53	37.31	98.72	61.59	59.79	75.16	69.78	81.03	66.45	49.78	63.37	79.66	65.77
SAM-Frozen	39.51	38.34	64.83	64.42	58.57	42.61	48.60	95.93	45.02	63.29	69.69	48.61	89.63	70.80	65.34	60.60	77.27	61.36
SAM-Adapter	56.87	**53.29**	67.31	68.72	68.43	57.39	55.77	**97.83**	63.85	67.33	75.63	66.92	**96.20**	73.64	67.04	70.82	86.02	70.18
SAM-LoRA	60.52	50.59	70.86	69.14	67.85	65.62	54.83	96.59	63.67	62.17	77.63	60.61	86.22	74.39	75.67	74.67	83.51	70.27
SAM-Full	27.65	39.78	43.54	39.71	49.38	31.57	15.45	90.59	51.36	30.90	58.39	51.09	88.22	54.26	28.86	52.39	61.42	47.92
Ours	66.31	52.49	**76.32**	70.38	69.57	76.80	**64.11**	**97.83**	**66.18**	**72.12**	**80.59**	73.43	95.16	**77.30**	76.38	**75.79**	**88.76**	**75.27**

Tables V and VI report the IoU of various methods for different categories for the WHDLD and DLRSD datasets. It can be observed that our method performs well in most of the categories for the WHDLD dataset. For instance, in the segmentation of bare soil, we achieved an IoU of 61.59%, which is approximately 0.6% higher than UNetFormer. More significantly, our method achieves the best or second-best results in all the categories for the DLRSD dataset, significantly outperforming the compared methods.

Furthermore, compared with other methods, our method exhibits more prominent enhancement in complex object segmentation. Specifically, in the WHDLD dataset, our method performs well in the building, road, pavement, and bare soil categories, with an average improvement of 4.5% and 4.2% compared with SAM-Adapter and SAM-LoRA, respectively. In contrast, for the vegetation and water categories with IoU exceeding 80%, our method only brings an average improvement of 1.8% and 2.0% compared with SAM-Adapter and SAM-LoRA, respectively. Similarly, in the DLRSD dataset, our method achieves significant improvements in most of the challenging categories, such as a nearly 9% improvement compared with o SAM-Adapter in the dock category, while exhibiting similar performance in the field and water categories. These results indicate that our method effectively reduces the loss of fine details and exhibits better segmentation performance, especially in handling complex objects.

Our method effectively works even when it deals with low spatial resolution datasets. Cloud detection in RS imagery has always been an important task. Therefore, we conducted

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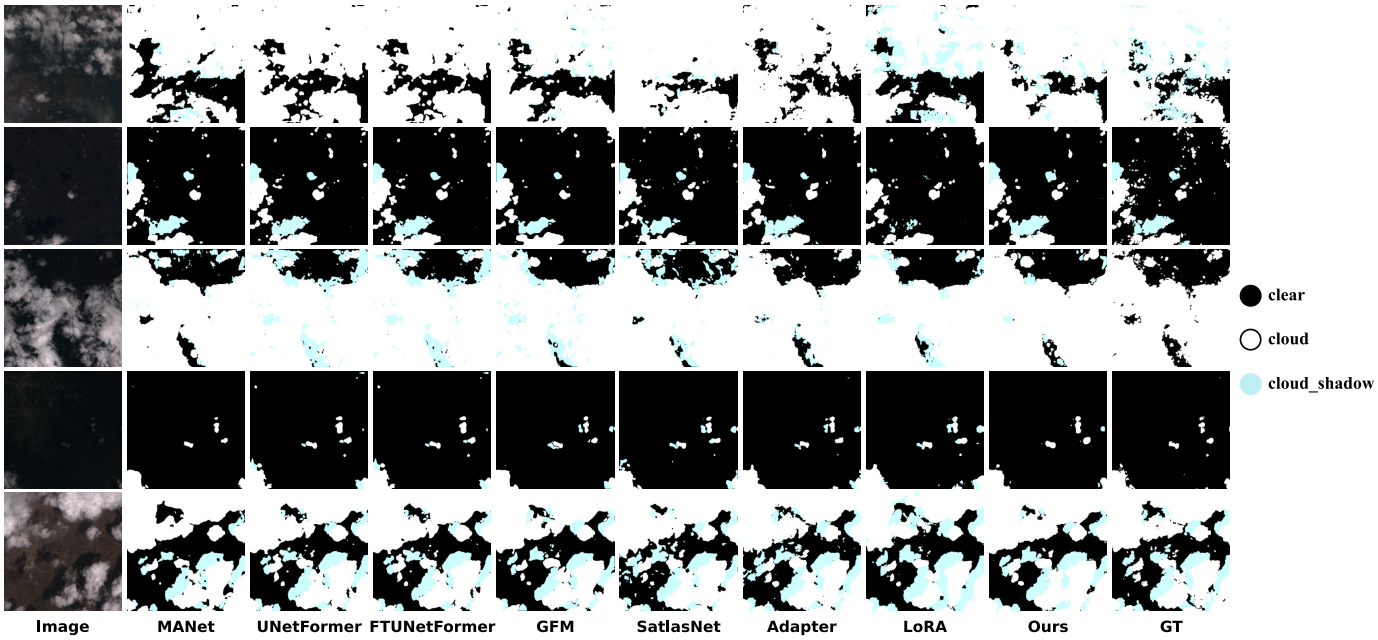


Fig. 8. Qualitative comparison of the visualization results of our method with other top-five methods and SAM-based methods on the Sentinel-2 Cloud Mask Catalogue dataset.

TABLE VI

IoU (%) per Class on the WHDLD Dataset. The Best Value in a Column Is Shown in Bold and the Second-Best Value Is Underlined

Method	building	road	pavement	vegetation	bare soil	water	**average**
SegNet	51.30	52.52	40.98	77.35	38.50	93.94	59.10
ABCNet	52.42	50.11	39.50	78.01	41.78	93.81	59.27
BANet	55.12	54.00	41.61	78.87	40.26	94.44	60.72
MANet	56.64	59.66	44.39	80.07	42.24	95.24	63.04
UNetFormer	56.22	58.86	44.64	79.36	<u>45.28</u>	94.67	63.17
MACUNet	58.11	59.23	45.12	**80.52**	42.35	**95.55**	63.48
FTUNetFormer	56.42	59.26	**46.65**	79.76	44.26	95.12	63.58
LOGCAN	**59.05**	<u>61.33</u>	46.07	<u>80.42</u>	43.88	<u>95.49</u>	<u>64.37</u>
Scale-MAE	44.84	47.51	39.49	74.42	39.25	89.12	55.77
GFM	51.22	46.99	35.82	74.38	40.07	90.82	56.55
SatlasNet	52.87	52.29	42.04	76.34	39.33	93.99	59.47
SAM-Frozen	34.78	41.40	32.67	65.58	17.01	85.64	46.18
SAM-Adapter	52.87	55.28	43.28	78.08	43.27	93.63	61.07
SAM-LoRA	55.11	57.07	41.59	78.08	42.13	93.08	61.18
SAM-Full	55.50	59.06	43.41	79.16	42.45	94.47	62.34
Ours	<u>58.68</u>	**61.59**	<u>46.45</u>	80.14	**45.84**	95.10	**64.63**

cloud segmentation experiments on the Sentinel-2 Cloud Mask Catalogue dataset, and the results are shown in Table VII. Compared with other methods, our approach has a clear advantage in cloud segmentation tasks, overcoming the advanced methods SAM-Adapter and SatlasNet by approximately 2% and 4%. Notably, the cloud_shadow class of pixels in the Sentinel-2 Cloud Mask Catalogue dataset accounts for only 1.5%, much less than the other two categories. Moreover, most of these pixels are located near cloud boundaries and contours, leading to shortcomings in identifying this category by most methods. In particular, both the SAM-Frozen and SAM-Full methods struggle with the cloud_shadow class, achieving the mIoU of approximately 0.001%. In comparison, our approach can more accurately segment this target. The visual results are shown in Fig. 8. Our method is clearly closer to real labels in the segmentation of many detailed contents, thus performing better than other methods.

2) Efficiency Analysis: To evaluate the application efficiency of our method, we calculate the number of parameters that need to be updated and the inference speed for all the methods. As shown in Table IV, although our methods are based on a foundation model, it does not incur significant computational costs. SAM-Adapter and SAM-LoRA achieved the smallest number of fine-tuning parameters, and the number of parameters that need to be updated in our method is much smaller than advanced models, such as LOGCAN and FTUNetFormer. Even though our method faces a disadvantage in terms of inference speed due to the complexity of the foundation model, the difference compared

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TABLE VII

QUANTITATIVE COMPARISON WITH SOTA MODELS ON THE SENTINEL-2 CLOUD MASK CATALOGUE DATASET. ON THE RIGHT IS IOU (%) PER CLASS. THE BEST VALUE IN A COLUMN IS SHOWN IN BOLD AND THE SECOND-BEST VALUE IS UNDERLINED

Method	OA(%)	F1(%)	mIoU(%)	clear	cloud	cloud_shadow
SegNet	89.80	76.75	65.81	80.17	83.93	33.32
ABCNet	91.26	73.18	63.84	83.56	85.99	21.98
BANet	87.83	75.57	63.86	78.48	79.69	33.39
MANet	91.26	80.61	70.19	83.23	85.95	41.38
UNetFormer	92.26	80.47	70.54	85.16	87.51	38.96
MACUNet	90.87	76.78	66.44	82.25	85.66	31.41
FTUNetFormer	92.08	79.90	69.97	84.55	87.91	37.46
LOGCAN	92.47	78.16	68.58	85.48	88.13	32.12
Scale-MAE	90.35	74.09	63.99	82.34	84.21	25.41
GFM	91.45	79.51	69.24	84.19	85.86	37.68
SatlasNet	91.32	80.60	70.20	83.80	85.61	41.20
SAM-Frozen	62.51	41.48	30.27	39.22	51.53	0.001
SAM-Adapter	93.40	81.45	72.10	87.57	89.15	39.58
SAM-LoRA	86.94	75.01	62.90	75.22	79.77	33.70
SAM-Full	87.86	58.98	52.86	77.67	80.89	0.001
Ours	**94.55**	**83.13**	**74.37**	**89.41**	**91.35**	**42.35**

TABLE VIII

QUANTITATIVE RESULTS FROM THE REPEATED EXPERIMENTS. ALL METRICS ARE REPORTED CALCULATING BOTH THE MEAN AND STANDARD DEVIATION FROM THE FIVE REPETITIONS. THE BEST VALUE IN A COLUMN IS SHOWN IN BOLD

Method	DLRSD		
	OA(%)	F1(%)	mIoU(%)
SegNet	78.27 ± 0.80	74.74 ± 1.44	61.70 ± 1.44
ABCNet	78.55 ± 0.48	75.08 ± 0.57	61.72 ± 0.64
BANet	77.86 ± 1.40	74.56 ± 1.54	61.78 ± 1.95
MANet	79.24 ± 0.44	76.92 ± 1.25	64.06 ± 1.40
UNetFormer	79.03 ± 0.60	77.80 ± 0.40	64.83 ± 0.60
MACUNet	81.83 ± 0.63	79.94 ± 1.51	67.85 ± 1.87
FTUNetFormer	80.78 ± 0.90	80.13 ± 0.49	67.92 ± 0.69
LOGCAN	82.11 ± 0.32	83.79 ± 0.60	72.95 ± 0.80
Scale-MAE	71.61 ± 0.35	64.77 ± 0.43	50.29 ± 0.47
GFM	67.28 ± 0.98	60.91 ± 1.63	45.99 ± 1.43
SatlasNet	79.41 ± 1.35	77.55 ± 0.98	64.46 ± 1.12
SAM-Frozen	75.57 ± 0.06	75.99 ± 1.14	62.61 ± 1.25
SAM-Adapter	83.09 ± 0.35	82.48 ± 0.38	70.91 ± 0.52
SAM-LoRA	83.14 ± 1.09	82.46 ± 0.91	70.92 ± 1.27
SAM-Full	69.44 ± 0.20	62.19 ± 0.56	47.59 ± 0.47
Ours	**84.76 ± 0.65**	**84.88 ± 0.41**	**74.71 ± 0.63**

with the existing models is not significant. In other words, our method achieves a significant improvement in segmentation performance without introducing excessive computational burden.

3) Repeated Experiments: To mitigate the randomness of experimental results and more convincingly demonstrate the effectiveness of our method, we conducted five repeated experiments on the DLRSD dataset. For each experiment, the training, validation, and testing datasets were repartitioned in a fixed ratio. The mean and standard deviation of the results from these repeated experiments are presented in Table VIII. It is evident that our method consistently outperforms the other approaches even when multiple experiments are performed.

4) High-Frequency Features Analysis: To further validate the effectiveness of MeSAM in extracting high-frequency information, we visualized the relative log amplitudes of the feature maps in the encoder after Fourier transform, as shown in Fig. 9. It can be observed that the features extracted by our method maintain a uniform distribution across different frequencies, with significantly higher intensity in high-frequency features compared with other methods.

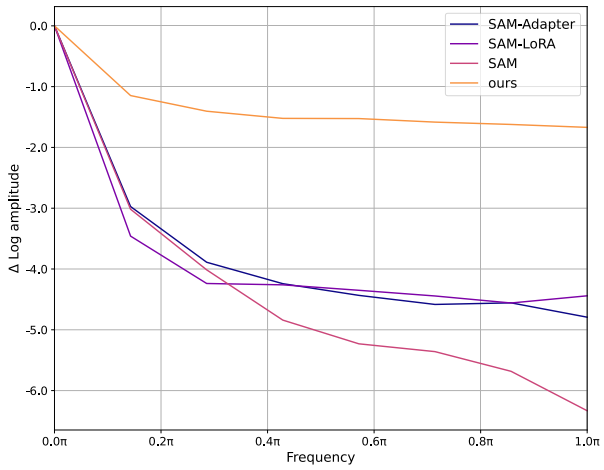


Fig. 9. Relative log amplitudes of the encoder feature maps of the different methods after Fourier transformation.

Among these methods, the image encoder in SAM extracts the least amount of high-frequency information, while the differences between SAM-Adapter and SAM-LoRA are relatively small.

5) Ablation Study: To investigate the contributions of each module in the proposed method, we conducted comprehensive ablation experiments on the DLRSD dataset. MeSAM can be divided into three components: inception former, RS correction, and multiscale connection. Therefore, we separately conducted experiments on these three modules.

The results are shown in Table IX, where ①, ②, and ③ refer to the inception former, the RS correction, and the multiscale connection components, respectively. Taking mIoU as an example, the experiments demonstrate that our

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TABLE IX
ABLATION STUDY ON THE DLRSD DATASET. ①, ②, AND ③ REFER TO THE INCEPTION FORMER, THE RS CORRECTION, AND THE MULTISCALE CONNECTION COMPONENTS

Method	Inc. Former	RS Cor.	Mul. Con.	OA(%)	F1(%)	mIoU(%)
SAM-Frozen				75.63	74.81	61.36
SAM-Full				69.58	62.58	47.92
SAM-Adapter				82.72	81.88	70.18
SAM-LoRA				82.81	82.02	70.27
SAM+①	✓			84.74	84.29	73.51
SAM+①+②	✓	✓		84.79	85.13	74.77
SAM+①+②+③	✓	✓	✓	**85.37**	**85.45**	**75.27**

method has achieved an improvement of approximately 5% compared with the baseline method. Specifically, around 3.3% improvement is attributed to the inception former module, over 1.2% improvement from the RS correction module, and the multiscale connection contributes to 0.5% improvement.

V. CONCLUSION

In this article, we explore an efficient paradigm of applying an image segmentation foundation model, e.g., SAM, to the task of semantic segmentation of RS images. Moreover, we propose MeSAM, which efficiently applies SAM to perform semantic segmentation on RS images. MeSAM aims to achieve end-to-end semantic segmentation of RS images based on SAM by fine-tuning a small number of parameters. It comprises two main modules: the Inception Transformer, designed for capturing high-frequency features, and the RS correction module, responsible for rectifying tokens fed into the mask decoder, enabling the mask decoder to fill the gap between natural images and RS images. We also incorporate multiscale connections between the image encoder and mask decoder to enhance the utilization of multiscale features, effectively recognizing objects of varying scales in RS images. The effectiveness of all the components of MeSAM is validated through ablation experiments. The experimental results on various datasets demonstrate that the proposed fine-tuning methods achieve much better results than many specialized models, and MeSAM obtains SOTA segmentation performance.

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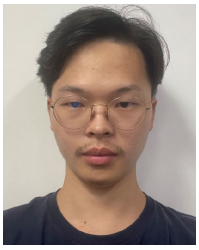
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